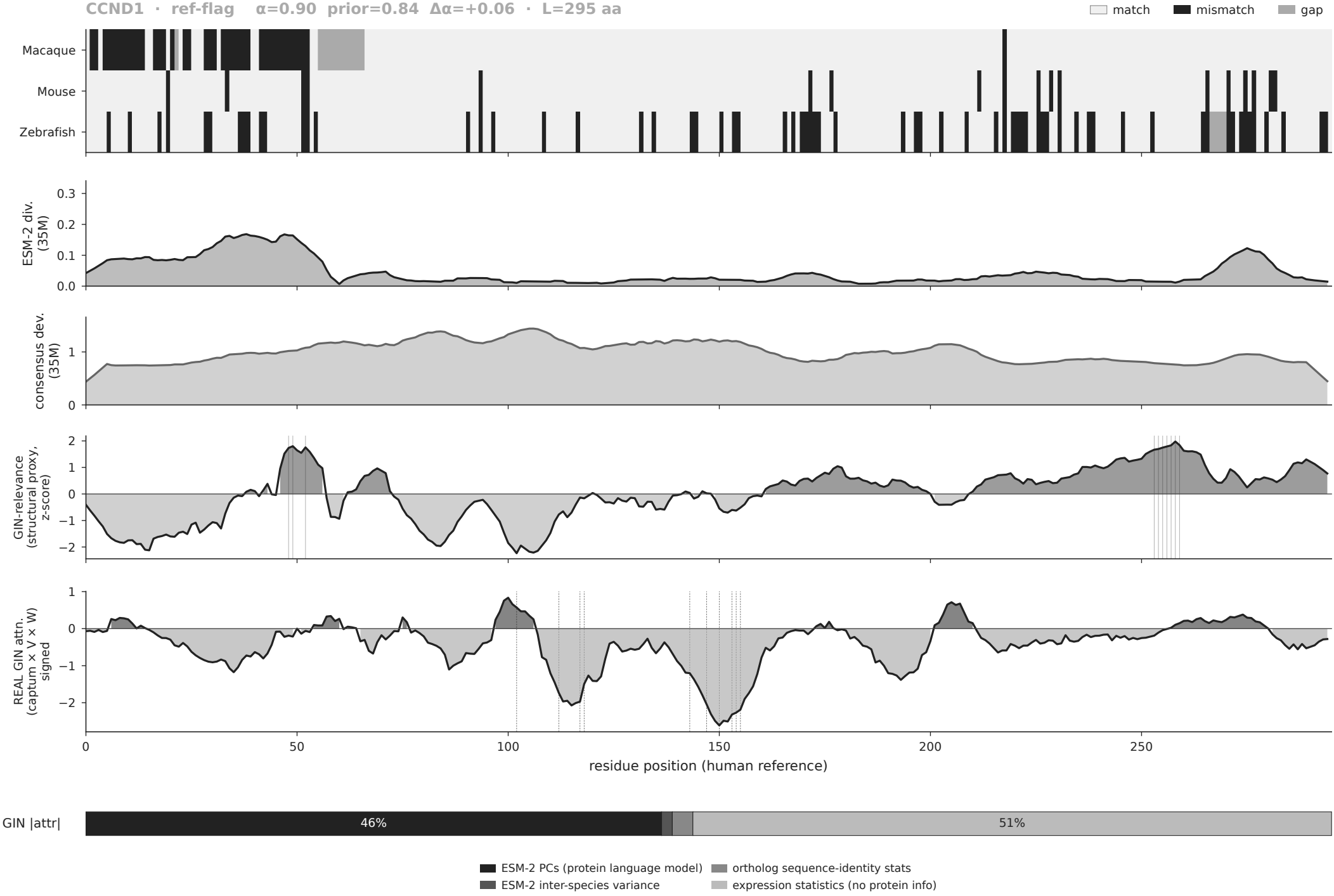
CCND1 · ref-flag α=0.90 prior=0.84 Δα=+0.06 · L=295 aa
match
mismatch
gap
Macaque
Mouse
Zebrafish
ESM-2 div. (35M)
0.3
0.2
0.1
0.0
consensus dev. (35M)
1
0
GIN-relevance (structural proxy, z-score)
2
1
0
−1
−2
REAL GIN attn. (captum × V × W) signed
1
0
−1
−2
0
50
100
150
200
250
residue position (human reference)
GIN |attr|
46%
51%
ESM-2 PCs (protein language model)
ESM-2 inter-species variance
ortholog sequence-identity stats
expression statistics (no protein info)